

Learning Day 6 Autoencoders

Project Summary with Visuals for the Three Completed Notebooks

This document summarises the three completed official Learning Day 6 notebooks: Dense Autoencoder, Convolutional Autoencoder, and Variational Autoencoder. The missing supplementary notebooks will be completed later, while their key theory is included in the Main Theory A-Z document.

How to read this document: The summary is written in simple language and focuses on what was done, why it was done, what the results mean, and how the visuals support the conclusions.

1. Completed Notebooks

Notebook	Level	Dataset	Main Concept	Status
LD6_Beginner_Dense_Autoencoder_MNIST.ipynb	Beginner	MNIST	Flattened image reconstruction using Dense layers	Completed
LD6_Intermediate_Conv_Autoencoder_FashionMNIST.ipynb	Intermediate	Fashion-MNIST	Image-aware reconstruction and feature extraction	Completed
LD6_Expert_Variational_Autoencoder_FashionMNIST.ipynb	Expert	Fashion-MNIST	Probability-based latent space and image generation	Completed

2. Key Results at a Glance

Experiment	Dataset	Architecture / Setup	Main Result	Interpretation
Dense AE on MNIST	MNIST	784 -> 128 -> 32 -> 128 -> 784	Avg MSE 0.008549; validation BCE about 0.0877	Good beginner reconstruction; loses sharp stroke details.
Conv AE on Fashion-MNIST	Fashion-MNIST	Conv2D + Pooling -> bottleneck -> UpSampling + Conv2D	Avg MSE 0.004471 on 2,000 test images	Preserves image structure better than dense layers.
Dense baseline on Fashion-MNIST	Fashion-MNIST	784 -> 128 -> 32 -> 128 -> 784	Avg MSE 0.022474 on 100 images	Baseline for fair Conv comparison.
Conv vs Dense comparison	Fashion-MNIST	Same 100 test images	Conv MSE 0.003979 vs Dense MSE 0.022474; 82.3% reduction	Conv AE is clearly better for image reconstruction.
VAE beta = 1	Fashion-MNIST	784 -> 128 -> z_mean/z_log_var/z -> decoder -> 784	Avg MSE 0.039027; KL 7.490874	Better reconstruction among the VAE models.
VAE beta = 4	Fashion-MNIST	Same VAE architecture with stronger KL term	Avg MSE 0.044348; KL 3.894046	More regularised latent space, blurrier reconstructions.

3. Notebook 1: Dense Autoencoder on MNIST

The first notebook introduced the core autoencoder idea. A handwritten digit image was flattened from 28 x 28 pixels into 784 values. The encoder compressed this input into a 32-dimensional bottleneck, and the decoder reconstructed the image.

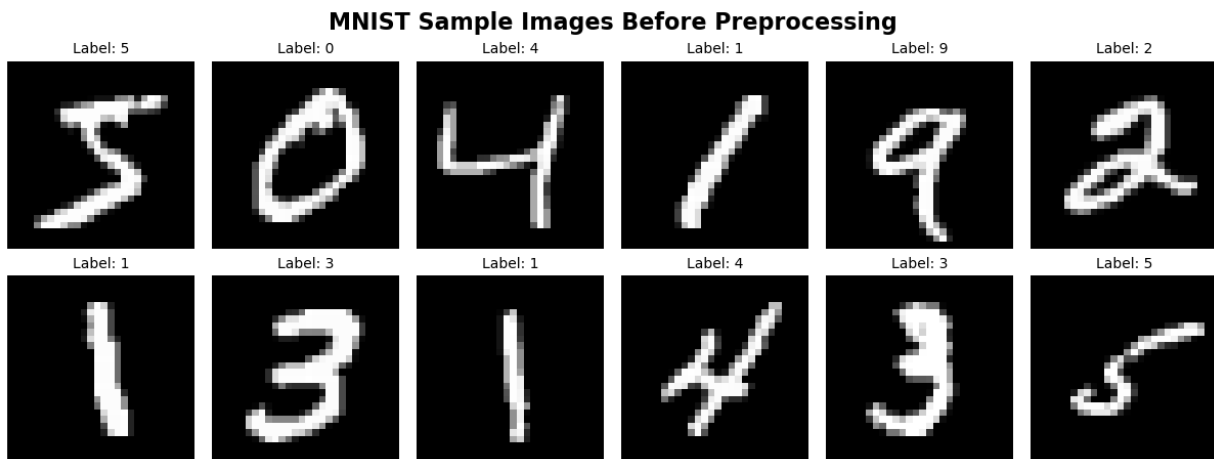


Figure 1: MNIST sample images before preprocessing.

Dense Autoencoder Flow

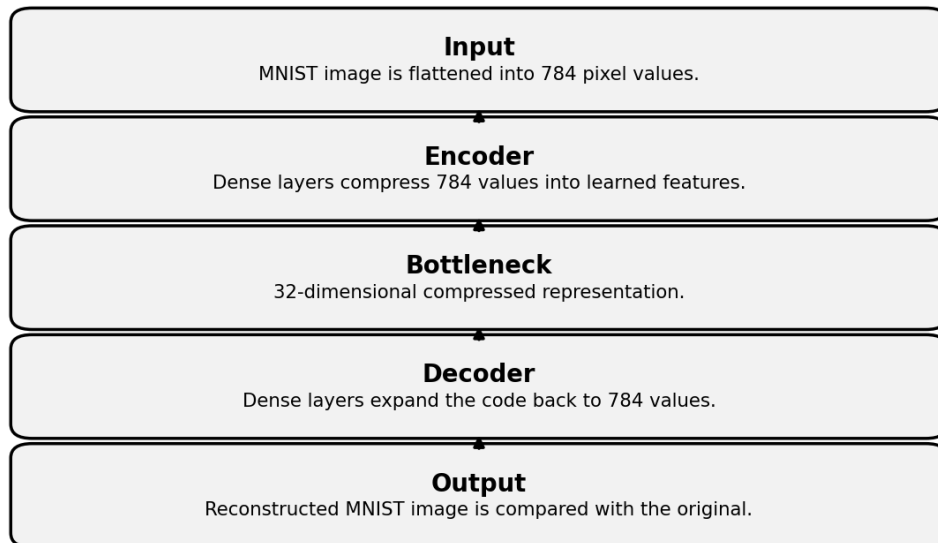


Figure 2: Dense Autoencoder workflow.

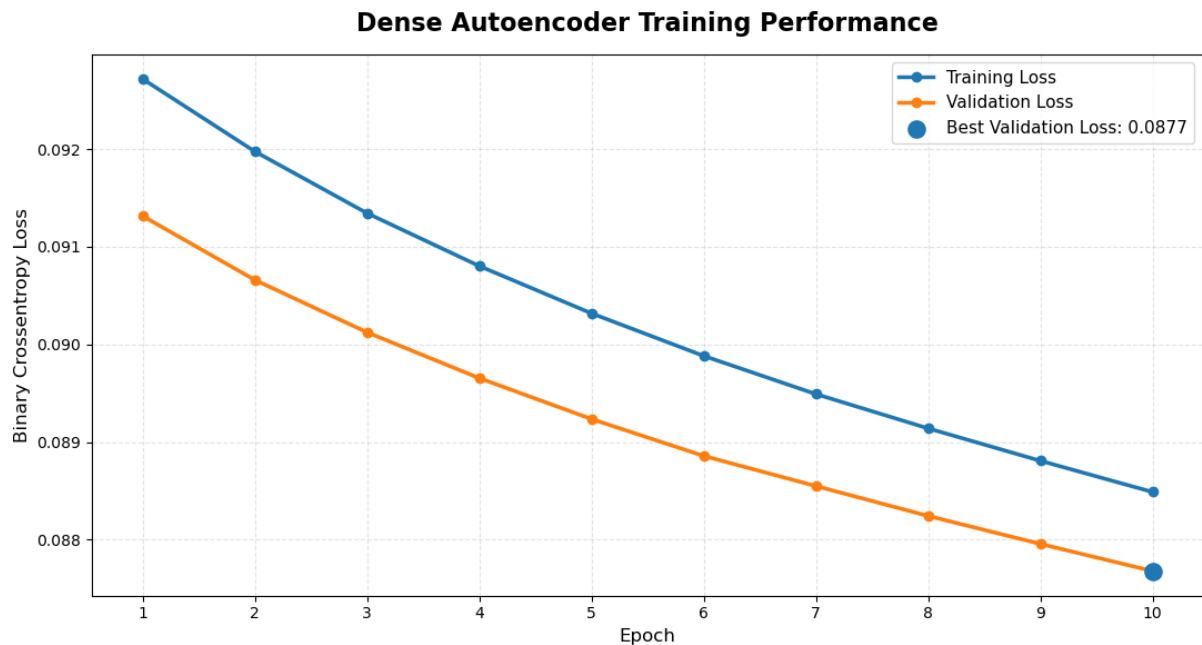


Figure 3: Dense Autoencoder training and validation loss curve.

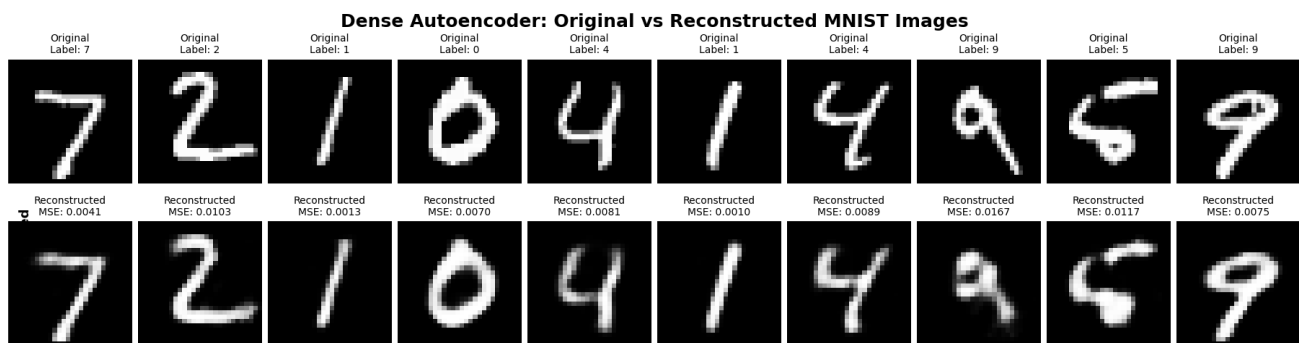


Figure 4: MNIST original vs reconstructed images.

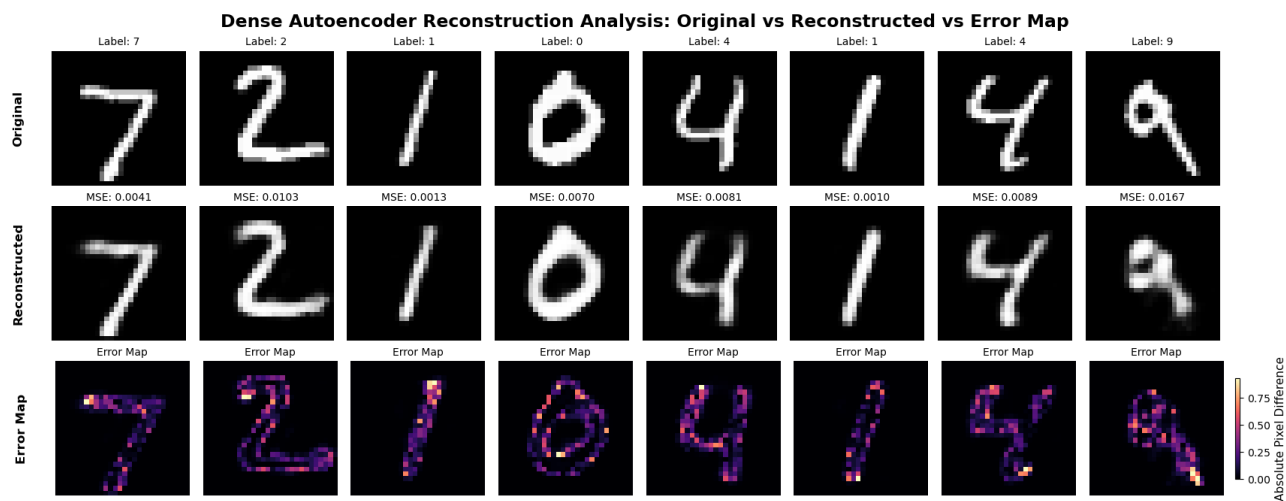


Figure 5: MNIST reconstruction error maps.

Key interpretation: the Dense Autoencoder preserved the main digit identity, but fine strokes, edges, and unusual handwriting shapes became smoother or blurrier. This is expected because the model compressed 784 pixels into only 32 values.

4. Notebook 2: Convolutional Autoencoder on Fashion-MNIST

The second notebook moved from flattened Dense layers to an image-aware Convolutional Autoencoder. Fashion-MNIST images contain object outlines, textures, sleeves, shoe soles, and clothing structure, so Conv2D layers are more suitable than only Dense layers.

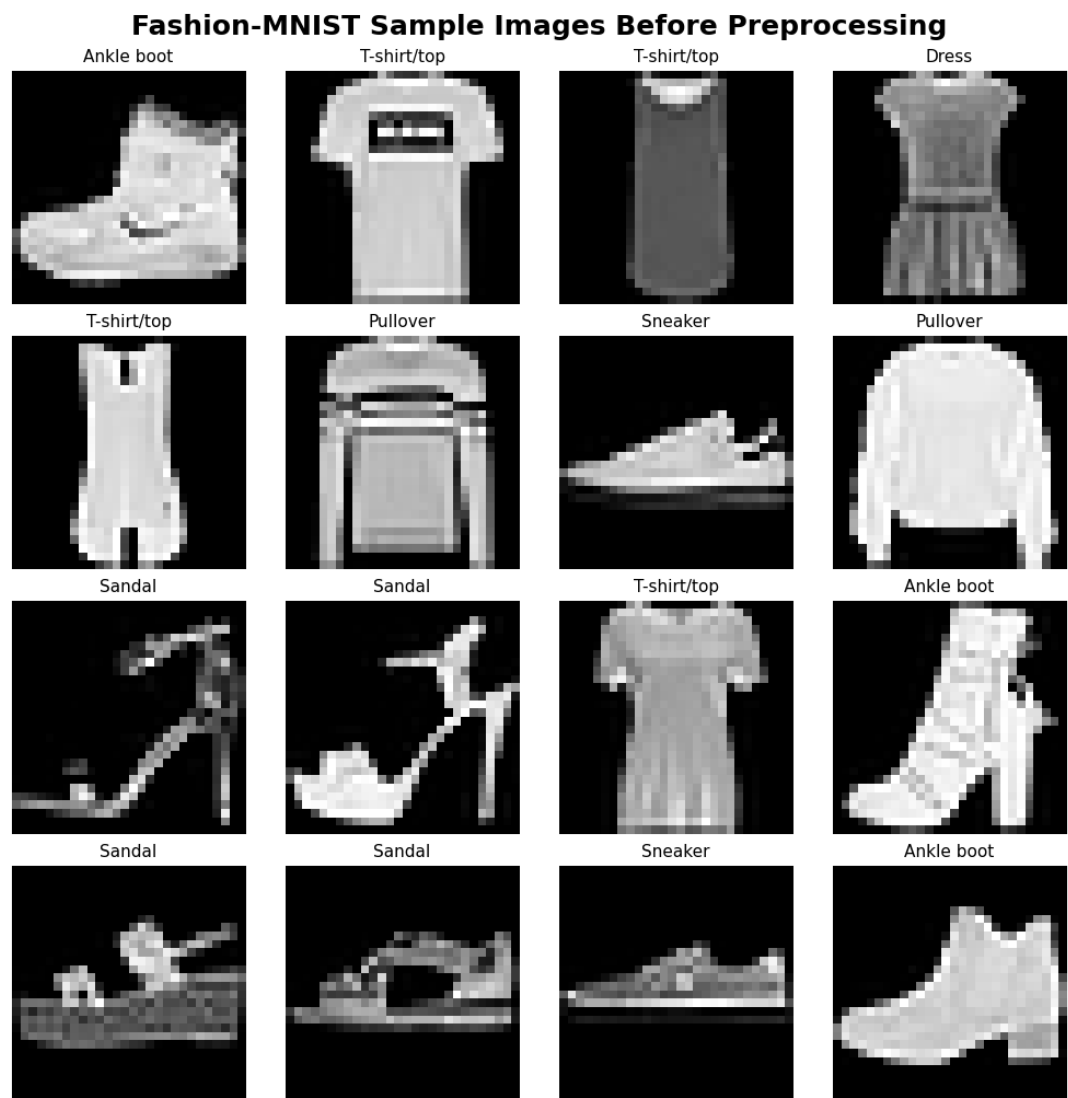


Figure 6: Fashion-MNIST sample images.

Convolutional Autoencoder Flow

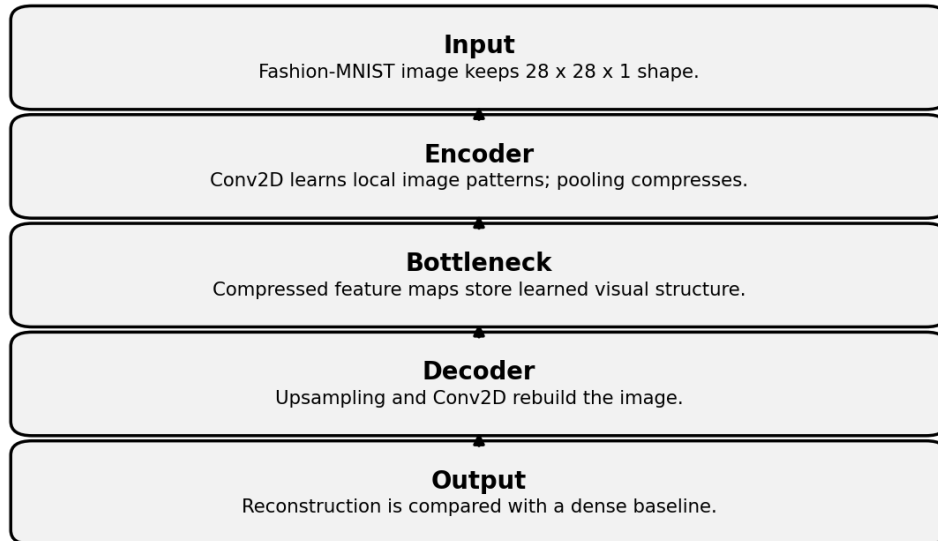


Figure 7: Convolutional Autoencoder workflow.

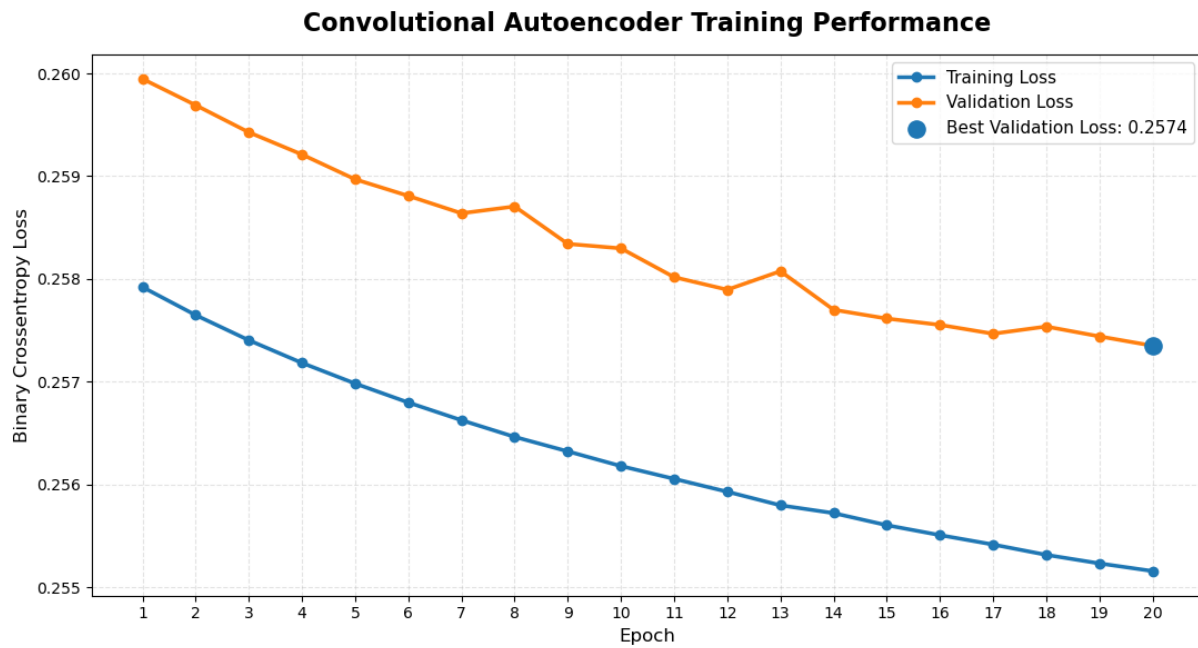


Figure 8: Conv Autoencoder training performance.

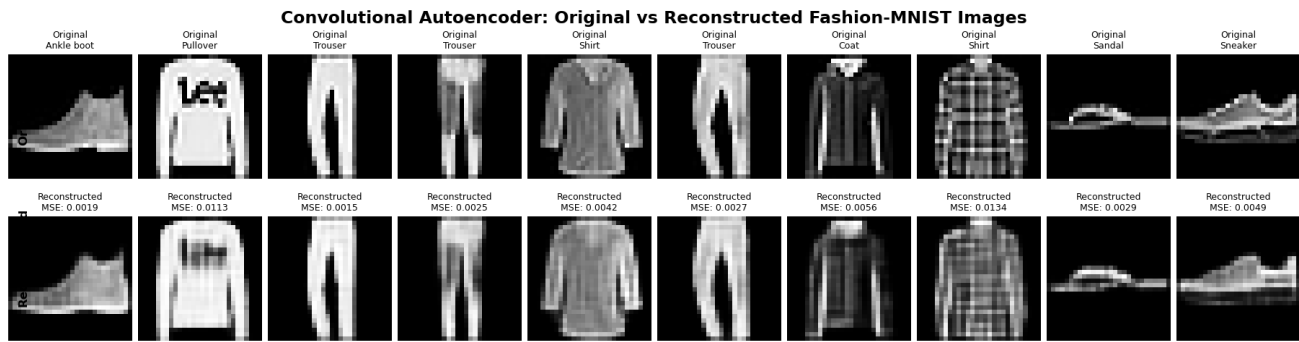


Figure 9: Fashion-MNIST original vs reconstructed images.

Model	Average MSE	Minimum MSE	Maximum MSE
Dense Autoencoder baseline	0.022474	0.008342	0.059023
Convolutional Autoencoder	0.003979	0.000969	0.019004

Key interpretation: the Conv Autoencoder reduced average reconstruction error by about 82.3% compared with the Dense baseline on the same 100 Fashion-MNIST test images. This shows that convolutional models are better for image reconstruction because they preserve spatial structure.

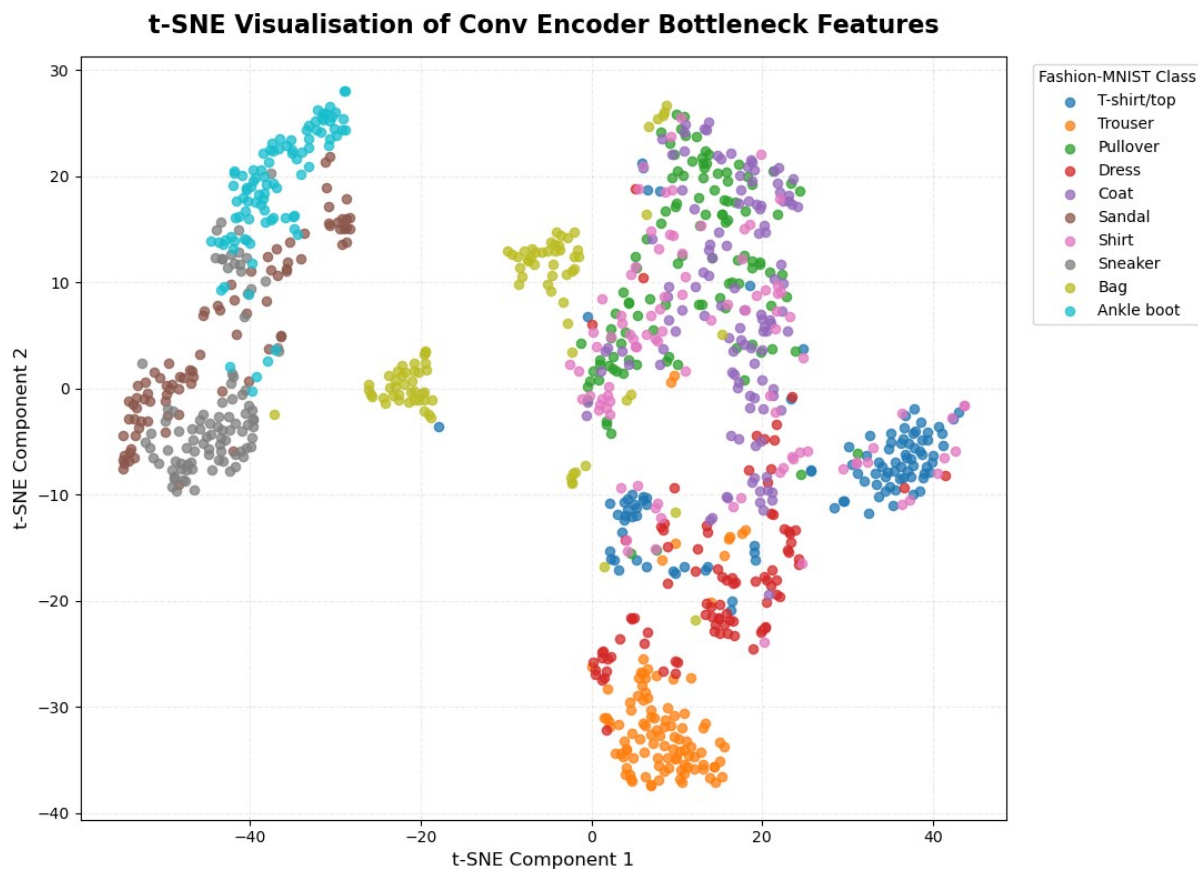


Figure 10: t-SNE visualisation of Conv Encoder bottleneck features.

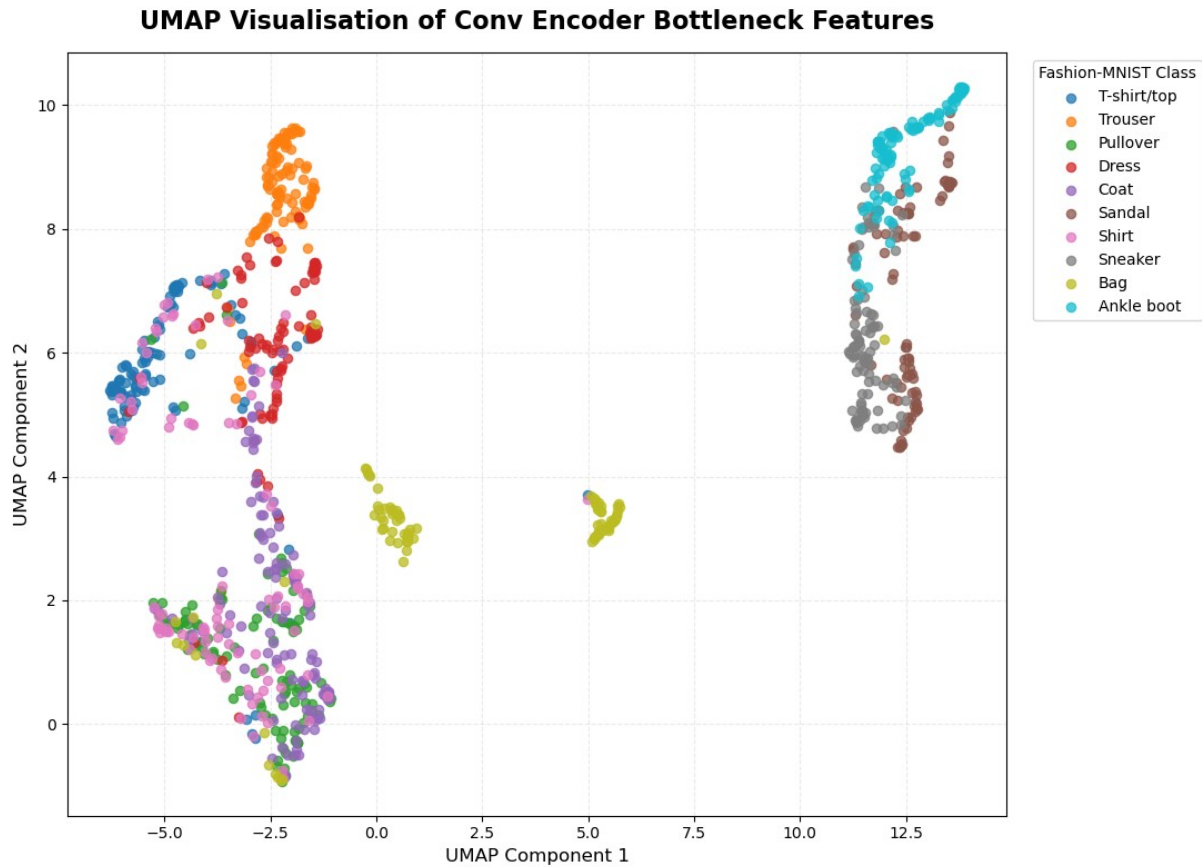


Figure 11: UMAP visualisation of Conv Encoder bottleneck features.

Feature insight: trousers, footwear, and bags showed clearer grouping. Upper-body clothing classes overlapped more because their shapes and textures are visually similar.

5. Notebook 3: Variational Autoencoder on Fashion-MNIST

The expert notebook implemented a VAE. Instead of producing one fixed bottleneck code, the encoder produced z_{mean} , $z_{\text{log_var}}$, and a sampled latent point z . The VAE was trained with reconstruction loss and KL divergence, using beta values 1 and 4.

Variational Autoencoder Flow

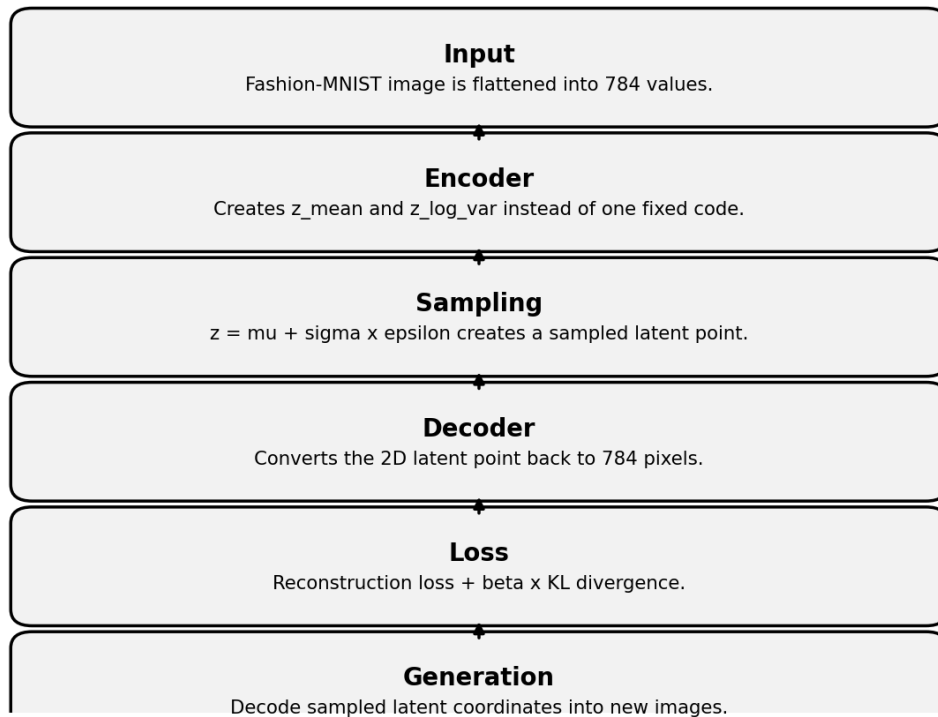


Figure 12: Variational Autoencoder workflow.

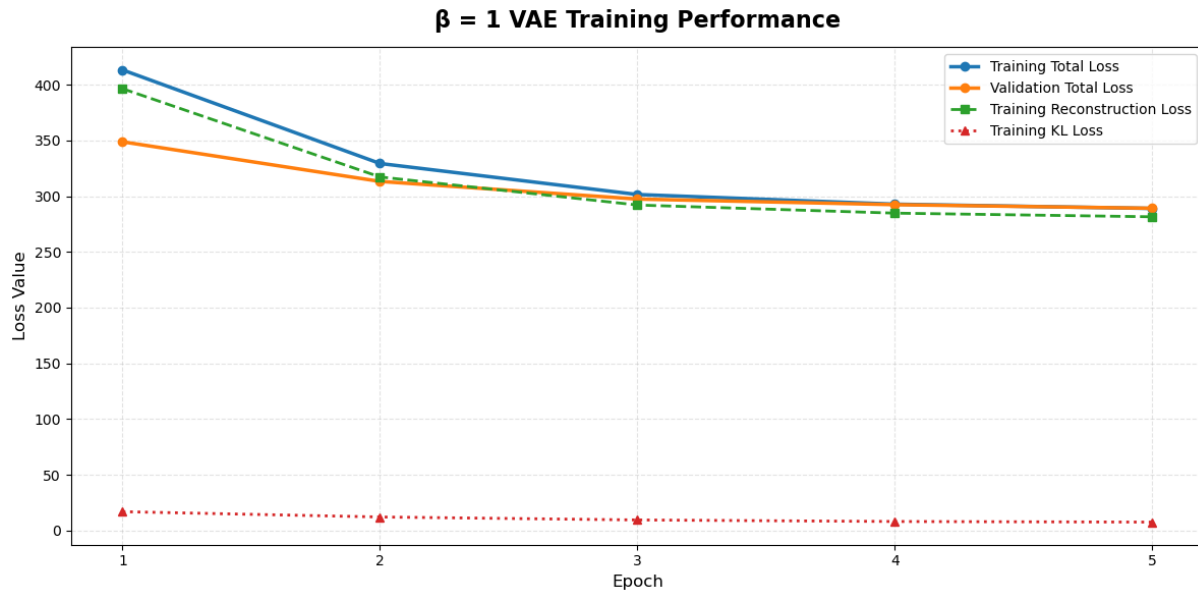


Figure 13: $\beta = 1$ VAE training performance.

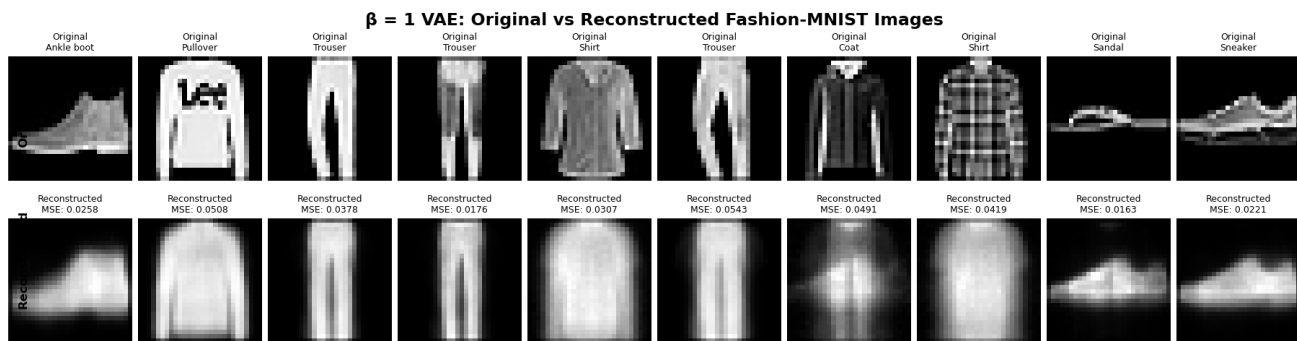


Figure 14: $\beta = 1$ VAE reconstructions.

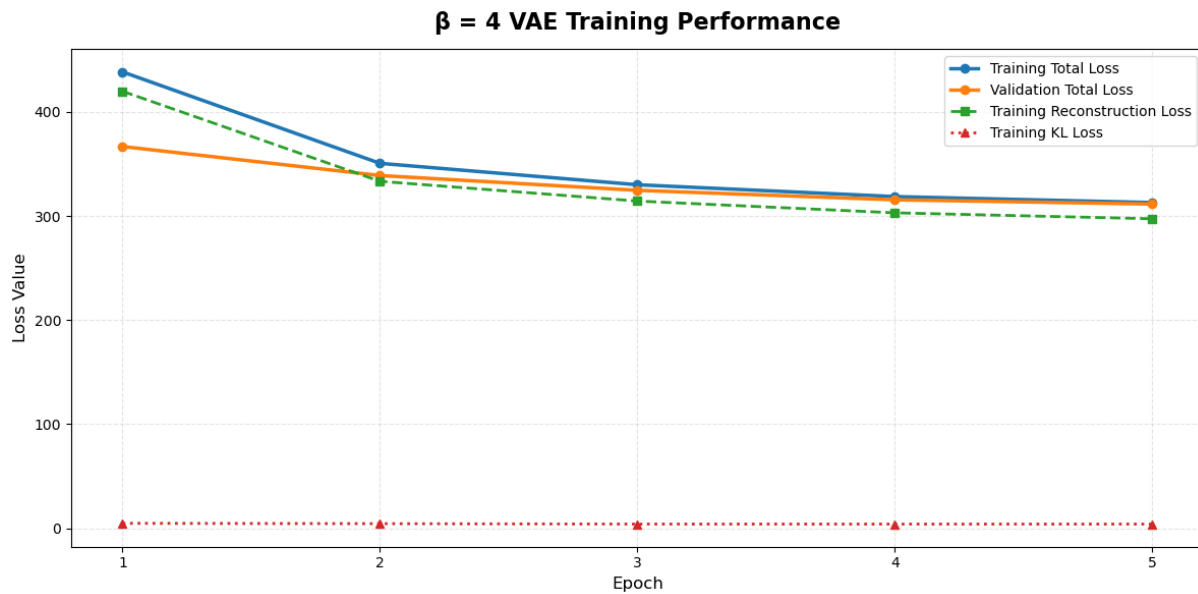


Figure 15: $\beta = 4$ VAE training performance.

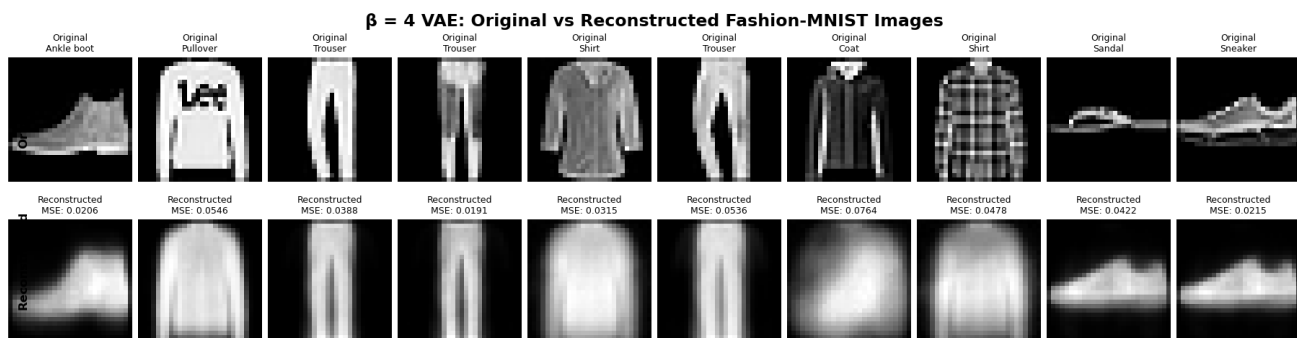


Figure 16: $\beta = 4$ VAE reconstructions.

Metric	VAE $\beta = 1$	VAE $\beta = 4$	Meaning
Average Test Reconstruction MSE	0.039027	0.044348	$\beta = 1$ reconstructs better
Final Train KL Loss	7.490874	3.894046	$\beta = 4$ regularises latent space more strongly
Final Train Reconstruction Loss	281.478516	297.320160	$\beta = 4$ sacrifices detail

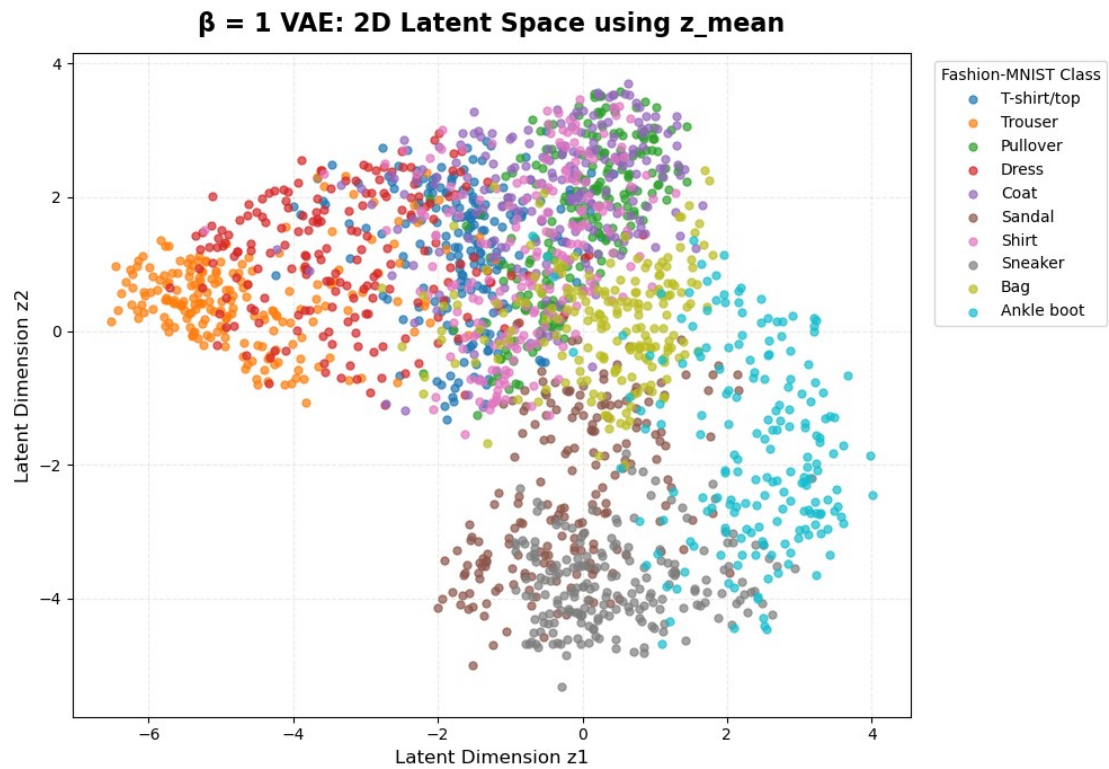


Figure 17: $\beta = 1$ latent space.

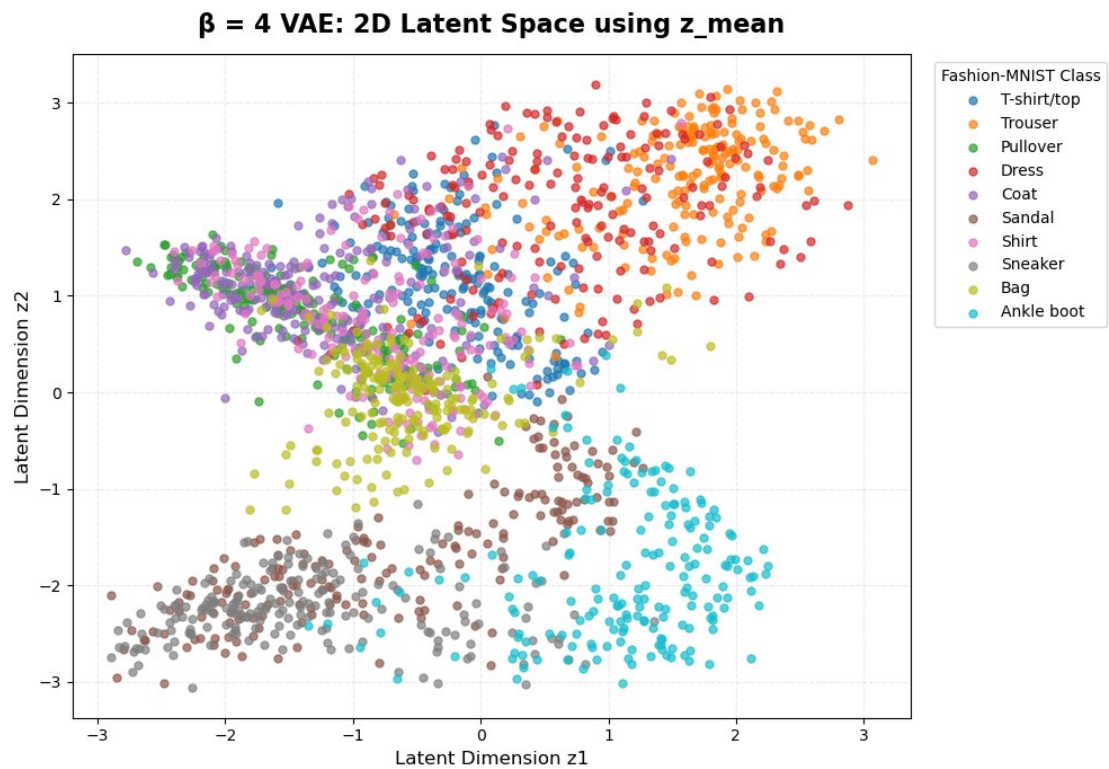


Figure 18: $\beta = 4$ latent space.

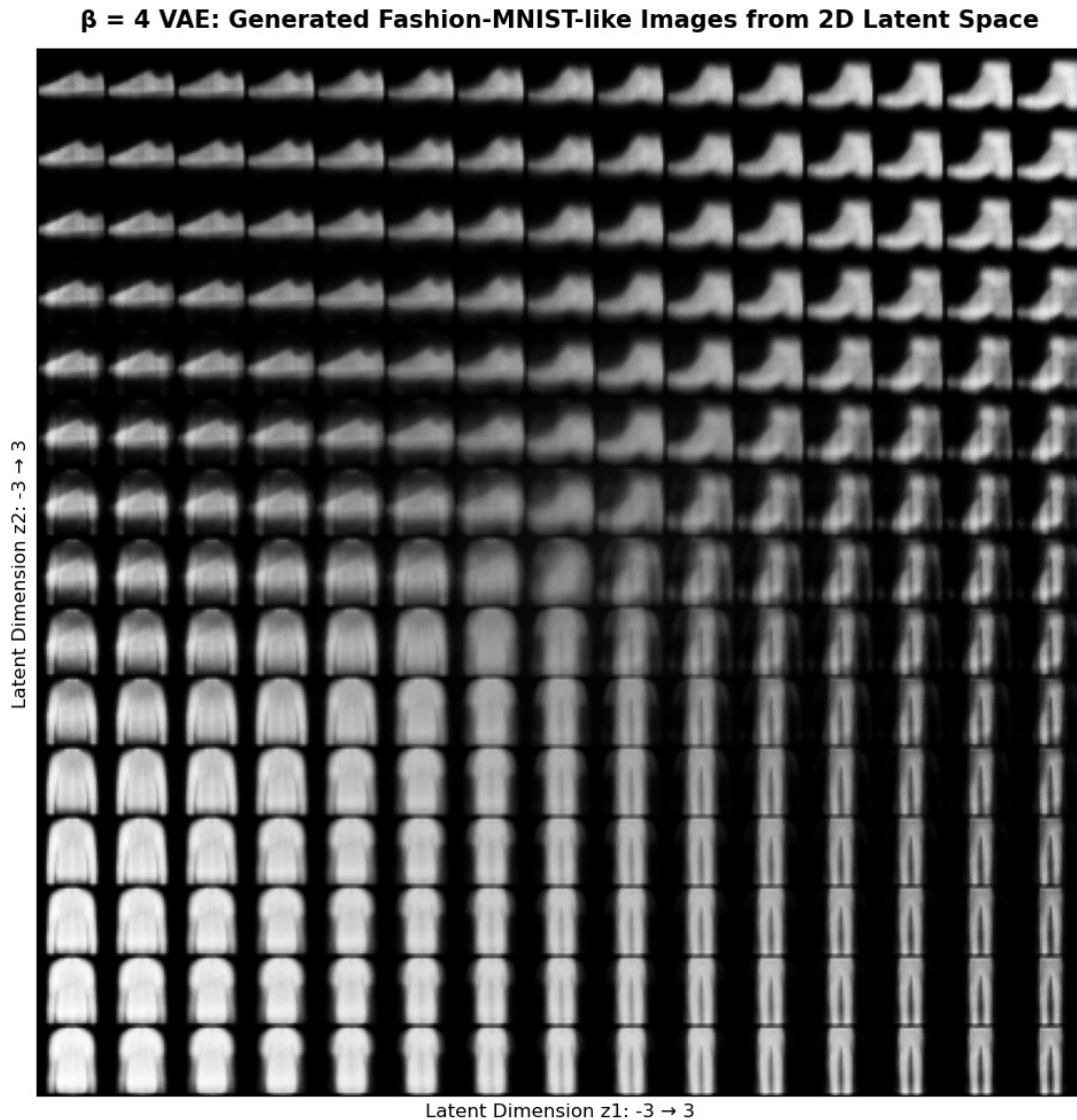


Figure 19: $\beta = 4$ generated image grid from 2D latent space.

Key interpretation: $\beta = 1$ gave better reconstruction quality. $\beta = 4$ pushed the latent space closer to a smoother normal distribution but produced blurrier reconstructions. The generated image grid confirms the generative ability of the VAE: moving through latent space gradually changes the generated fashion-like images.

6. Final Project Takeaways

- Dense Autoencoders are good for learning the basic compression-reconstruction idea.
- Convolutional Autoencoders are better for image data because they preserve spatial structure.
- Encoder bottleneck features can be used for feature extraction and visualisation.
- VAEs are generative models because they learn a smooth probability-based latent space.
- Increasing β strengthens latent-space regularisation but usually reduces reconstruction detail.

7. Source Note

Source basis: the three uploaded LD6 notebooks, the educx/MVD Learning Day 6 exercise collection, and the tutor Autoencoder PDF/PPT materials.